

Computer Vision and Image Processing

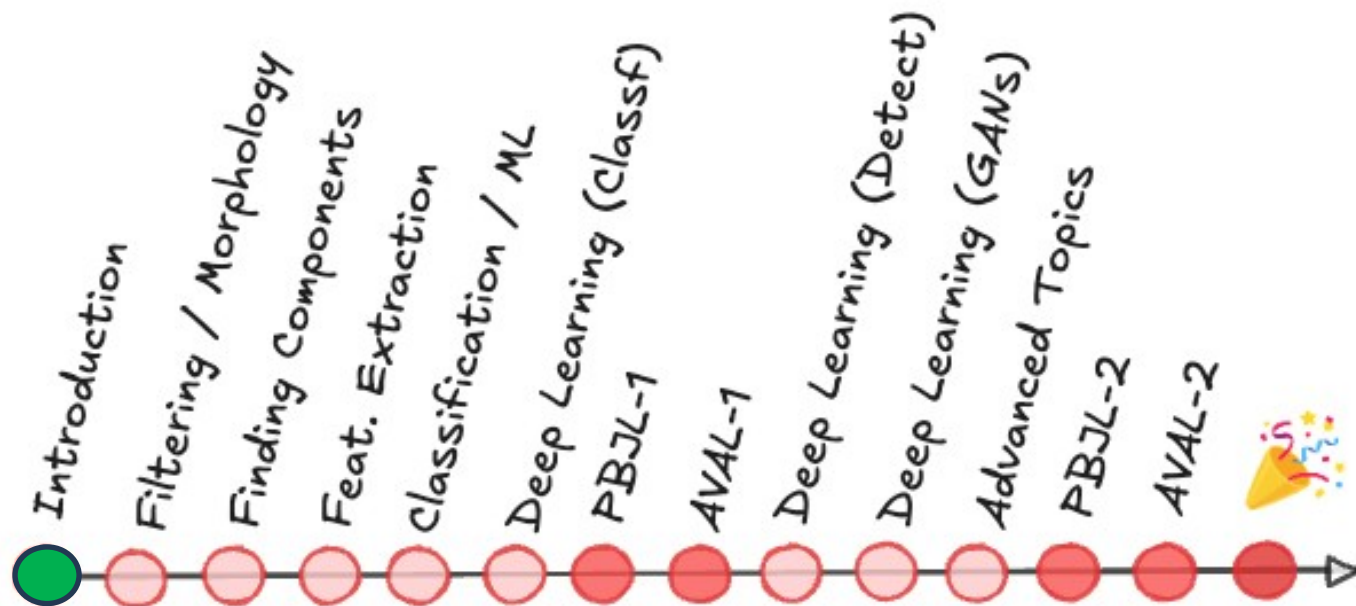
Prof. André Gustavo Hochuli

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aghochuli@ppgia.pucpr.br

Topics

- Apresentação do Professor
- O que esperar da disciplina?
- Ferramentas do estado da arte
 - Google Colab
 - OpenCV
 - Tensorflow
 -
- Exercícios

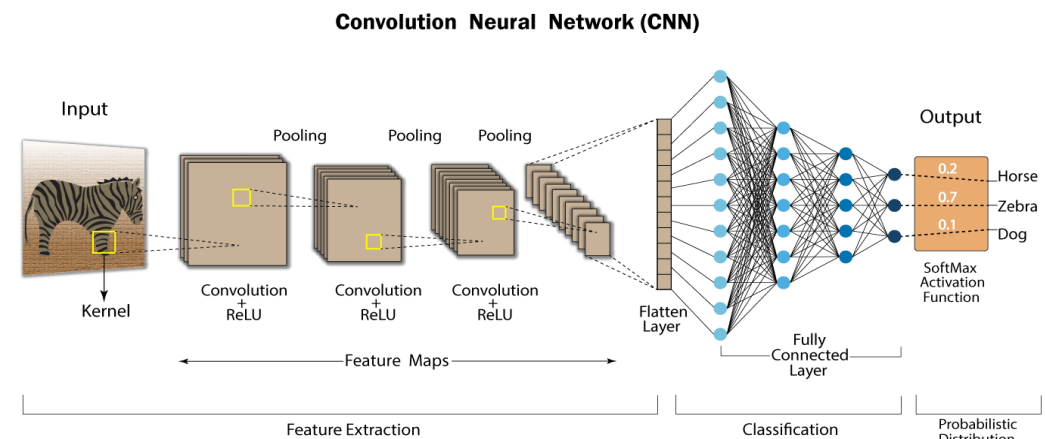


Prof. André Gustavo Hochuli

- Formação
 - Ciência da Computação [2004, PUCPR]
 - Mestre [2007, PPGIA/PUCPR]
 - Doutor [2018, PPGINF/UFPR]
- Experiência Profissional
 - P&D em Visão Computacional [2008-2013]
 - Professor Universitário [2014 – Atual]
- Linhas de Pesquisa
 - Aprendizagem de Máquina e Reconhecimento de Padrões

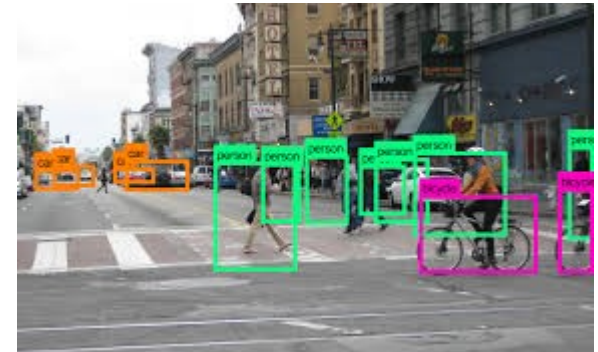


Hobbies:
Aviação
Futebol
Tecnologia



O que esperar da disciplina ?

- Processamento de Imagens
- Aprendizagem de Máquina
- Resolução de problemas
- Desafios encontrados no cotidiano
- Aulas teóricas e práticas
- Espaço para o estudante debater e trazer problemas/dúvidas
- Conteúdo incremental
- Trabalhos práticos
- Provas práticas
- OBS: Material didático em língua inglesa.



Plano de Ensino (Resumo)

Resultados de aprendizagem
RA1. Compreender fundamentos teóricos e práticos de Visão Computacional
RA2. Implementar soluções práticas em Visão Computacional

Resultado de Aprendizagem (RA)	Peso no RA				Nota do RA (0 a 10)
	Prova 1	Trabalho 1	Prova 2	Trabalho 2	
RA1	50%	–	50%	–	Soma ponderada das notas obtidas nas atividades
RA2	–	50%	–	50%	Soma ponderada das notas obtidas nas atividades

- A nota semestral é calculada a partir da ponderação dos dois RAs, conforme a seguinte fórmula:

$$\text{Nota semestral} = (\text{RA1} \times 0,7) + (\text{RA2} \times 0,3)$$

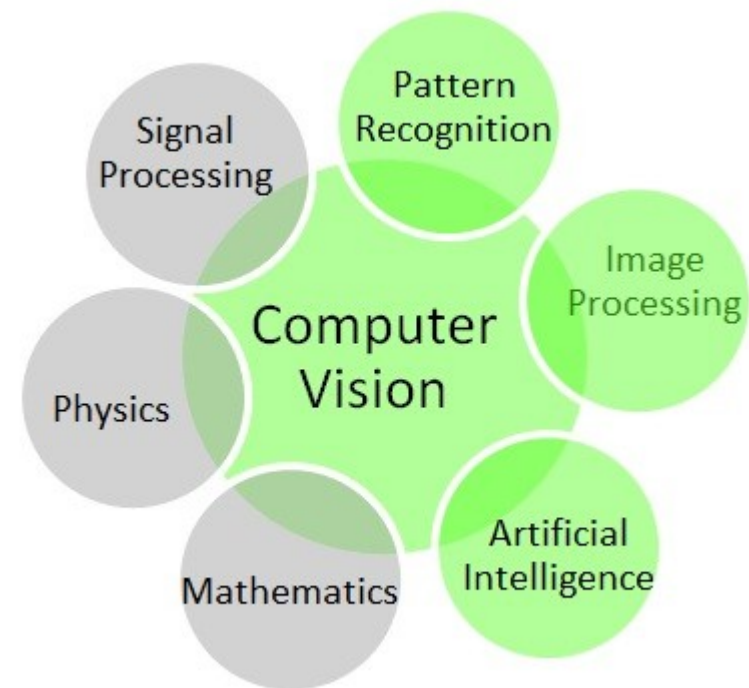
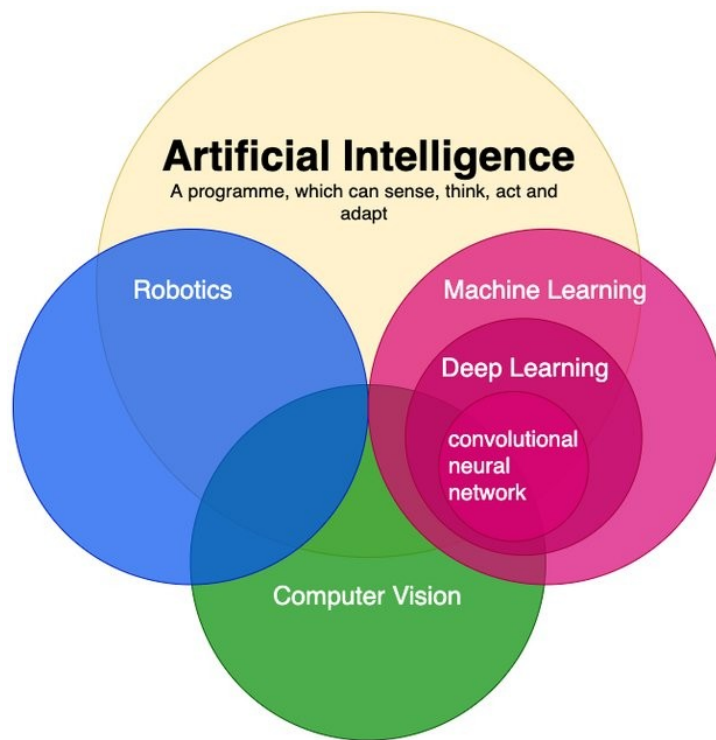
- A Recuperação Parcial será ofertada exclusivamente para o RA1;

- PLANO DE ENSINO COMPLETO DISPONIBILIZADO NO (CANVAS)

Plano de Ensino (Resumo)

Data	RA	Descrição das atividades pedagógicas	Carga horária
07/08/2026	RA1 e RA2	Apresentação do plano de ensino, ementa, formato de avaliação. Definições básicas (Visão e Representação de Imagens).	4h/a
14/08/2026	RA1 e RA2	Filtragem de imagens.	4h/a
21/08/2026	RA1 e RA2	Morfologia Matemática.	4h/a
28/08/2026	RA1 e RA2	Segmentação de Componentes.	4h/a
04/09/2026	RA1 e RA2	Extração de Características e Reconhecimento de Padrões.	4h/a
11/09/2026	RA1	Prova 1.	4h/a
18/09/2026	RA2	Feedback da Prova 1 e Apresentação de Trabalho	4h/a
25/09/2026	RA1 e RA2	Introdução ao Aprendizado Profundo. Redes Convolucionais Neurais (CNNs).	4h/a
02/10/2026	RA1 e RA2	Overfitting, Ajuste de Hiperparâmetros, Função de Perda, Regularização, Data Augmentation, Transfer Learning, Fine-Tuning.	4h/a
09/10/2026	RA1 e RA2		4h/a
16/10/2026	RA1 e RA2	Detecção de Objetos. Segmentação de Imagens.	4h/a
23/10/2026	RA1 e RA2		4h/a
30/10/2026	RA1 e RA2	IA Generativa.	4h/a
06/11/2026	RA2	Apresentação de Trabalho	4h/a
13/11/2026	RA1	Prova 2.	4h/a
20/11/2026	-	Feriado Nacional - Consciência Negra	-
27/11/2026	RA1	Recuperação parcial (prova), de acordo com a necessidade de cada estudante.	4h/a
04/12/2026	RA1	Recuperação estendida (prova), de acordo com a necessidade de cada estudante.	-

What is Computer Vision (CV) ?

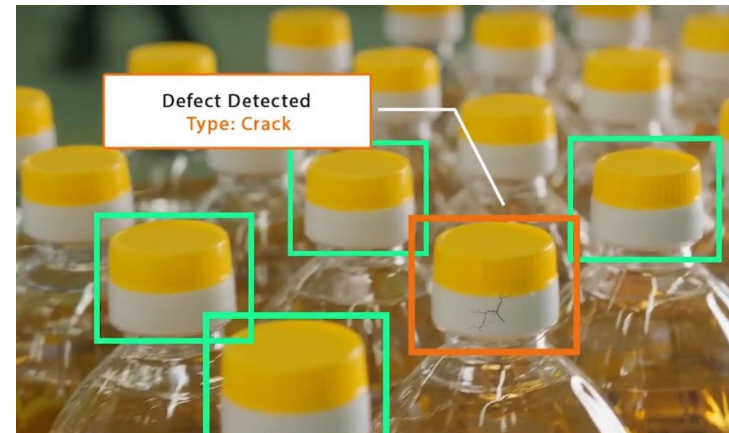
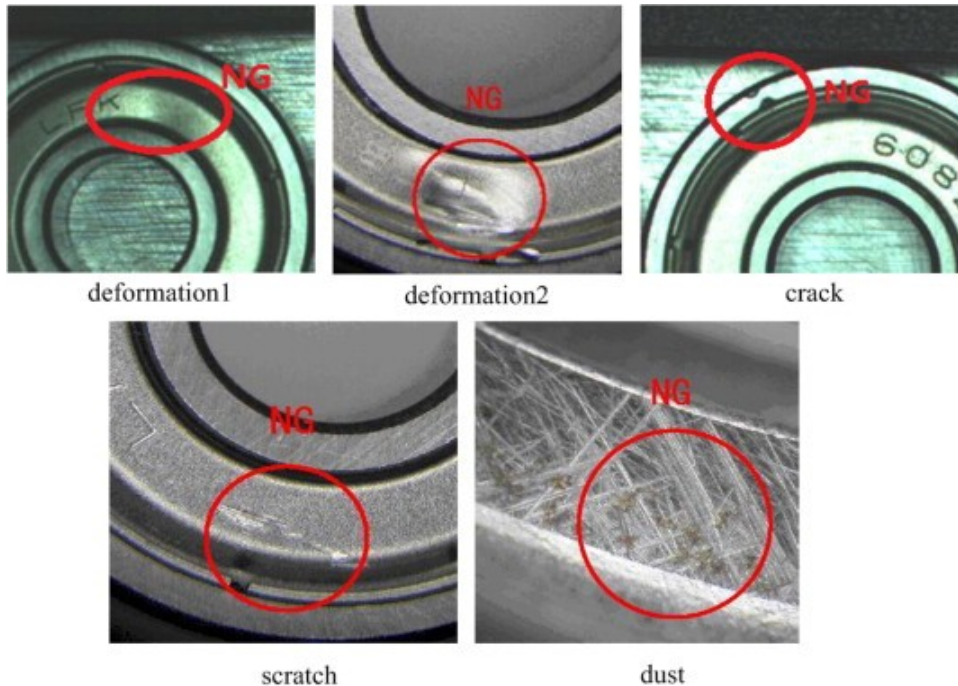


Tools & Libraries



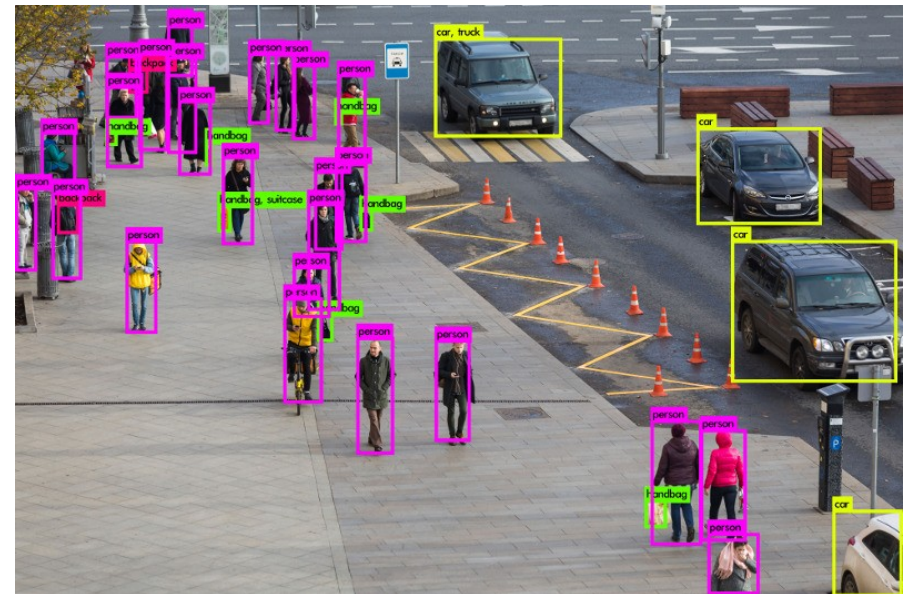
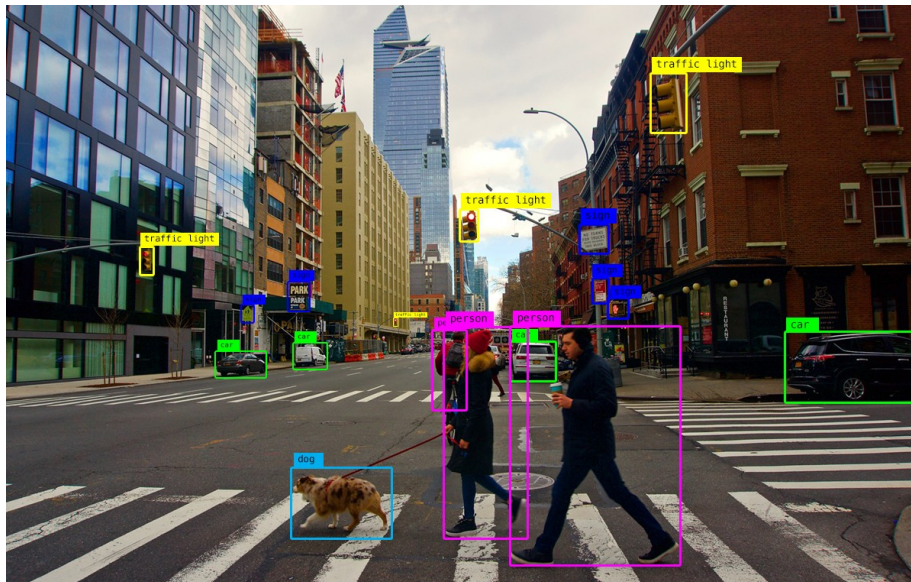
Computer Vision Applications

- Defect Detection and Quality Assurance



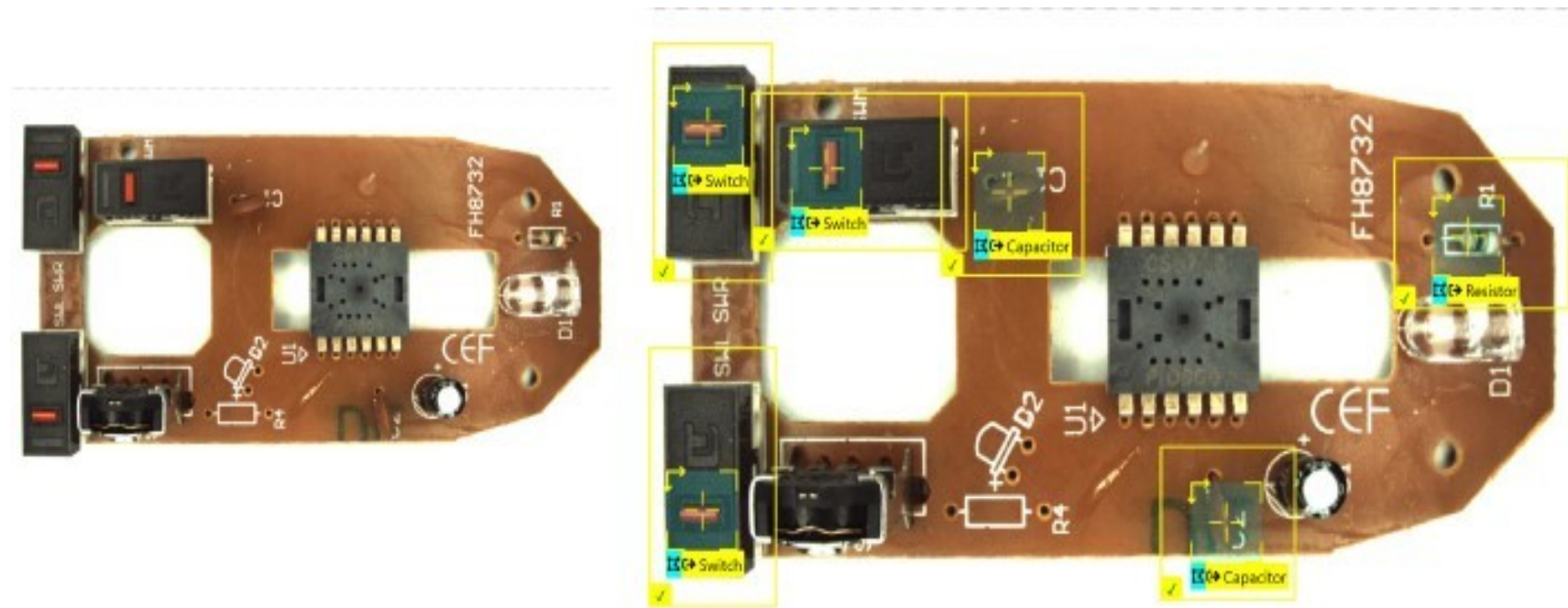
Computer Vision Applications

- Video Surveillance and Analytics



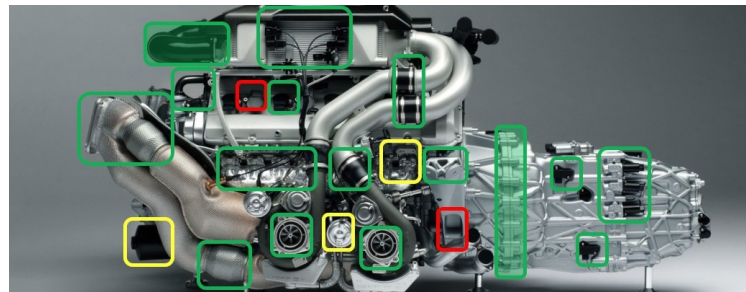
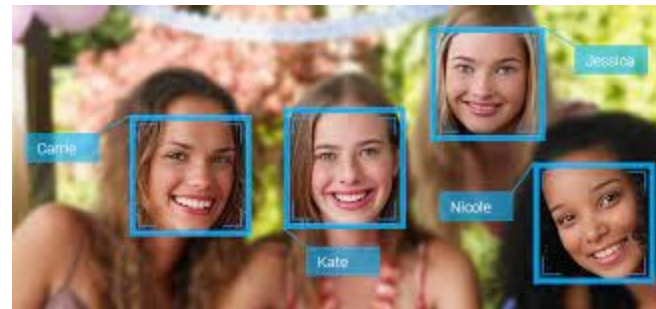
Computer Vision Applications

- **Assembly Verification**



Computer Vision Applications

- ...and....



Jobs

- **This is a field with several projects and open positions around the world.**
 - **Linkedin**
 - **Glassdoor**

Image Processing

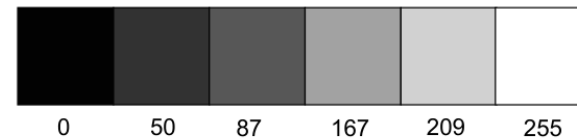
Basic Definitions

- **Basic Definitions**
 - **Pixel, Resolution and Scale**
- **Color Systems**
 - **GrayScale, RGB | BGR, CYMK, HSV**
 - **Color Conversion**
- **Binarization and Image Filtering**
- **Exercises**

Basic Definitions

Image and Pixel

- Image: Matrix NxM
- Pixel The smallest information of an image
- Range from 0 (black) to 255 (white)



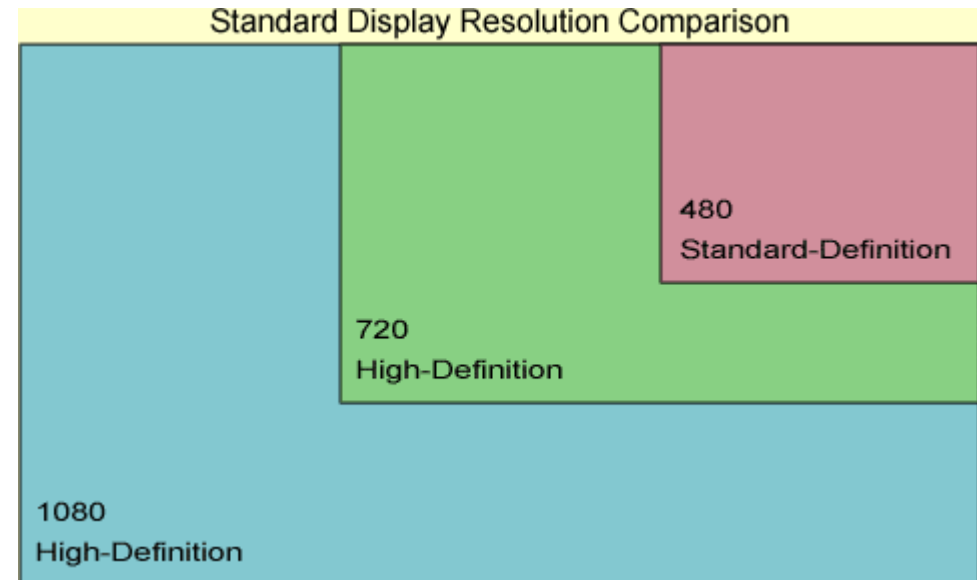
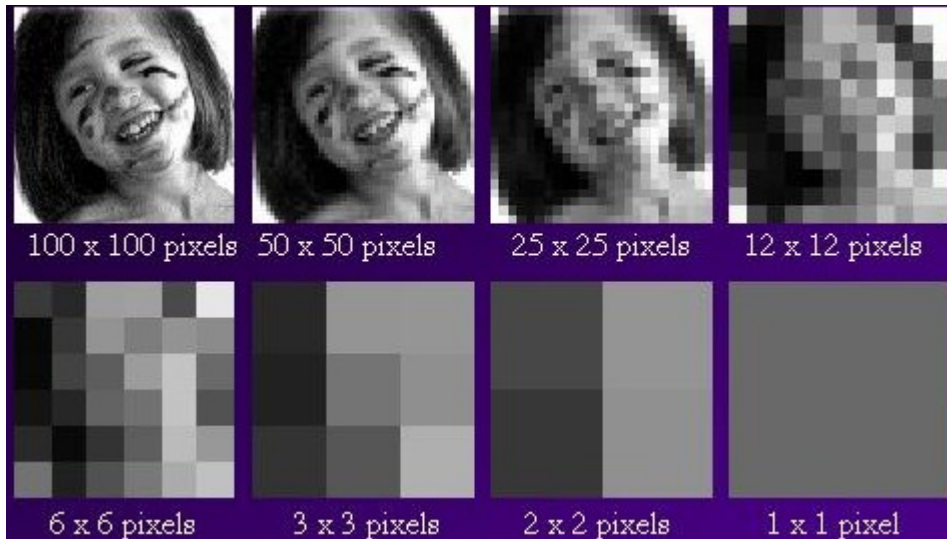
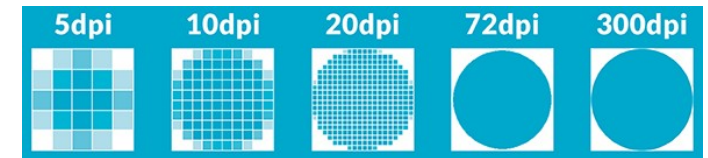
157	153	174	168	150	162	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	85	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

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155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	85	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
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183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

Basic Definitions

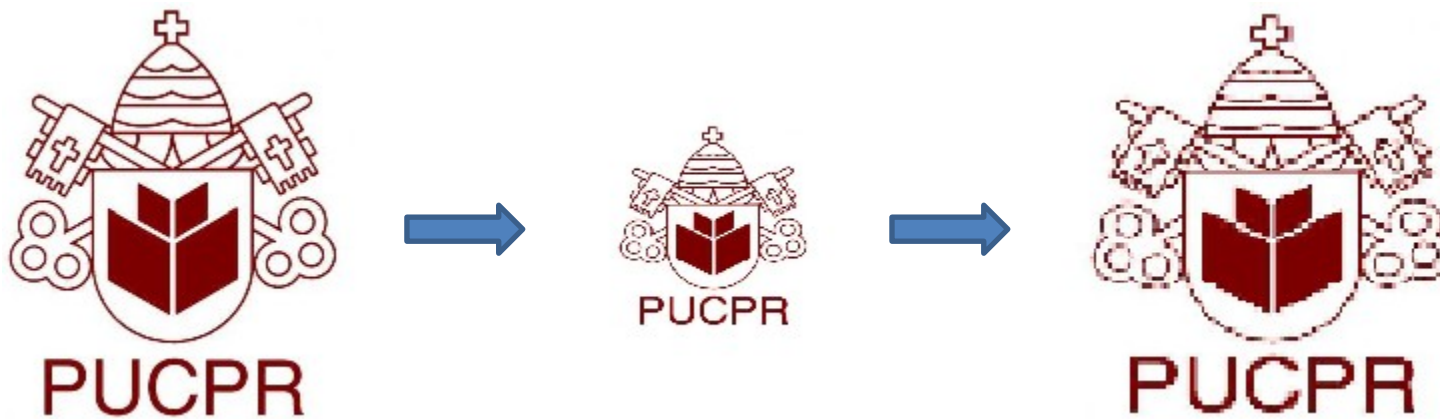
Image Resolution

- Pixel per Inch (PPI) for digital devices
- Dots per Inch (DPI) for analog devices (printers)
- Range from 0 (black) to 255 (white)



Basic Definitions

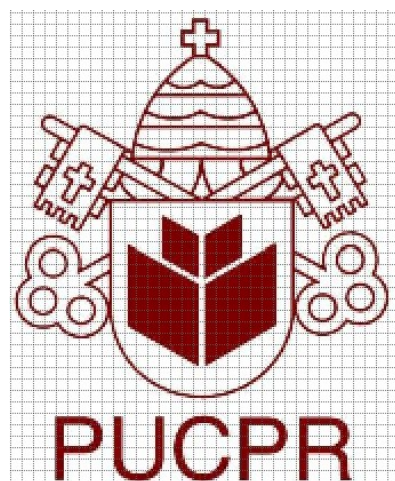
What is the problem?



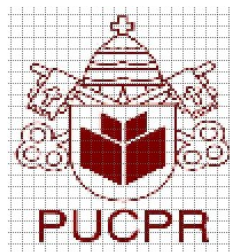
Basic Definitions

Downscale and Upscale (Resize)

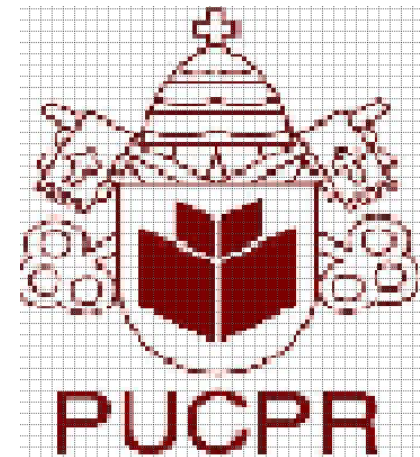
- Resolution loss
- Interpolation



160x200



96x106



160x200

Basic Definitions

ColorSpaces

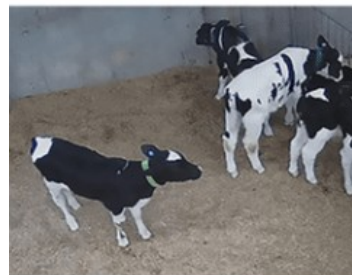
- **Binary (0-1) – 1 Channel**
- **GrayScale (0-255) – 1 Channel**
- **Color RGB – 3 Channels**
(Red, Green, Blue)Color



1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	0	0	1	1
1	1	1	1	1	1	0	0	1	0
1	1	1	1	1	1	0	1	1	0
1	0	1	1	0	1	1	1	0	0
1	0	0	0	0	0	1	1	1	1
1	0	0	0	0	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1



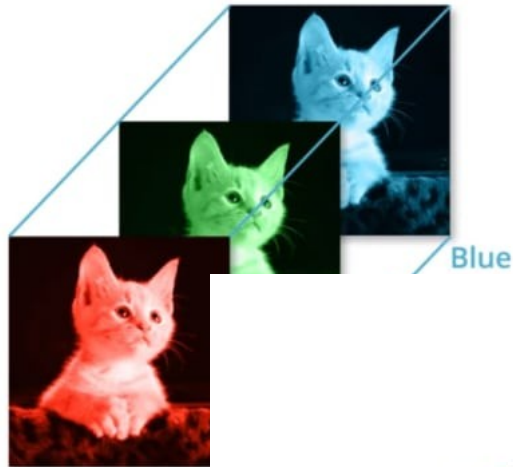
123	123	123	123	123	123	100	110	120	123
123	123	123	123	130	120	35	30	100	110
123	123	123	110	110	100	35	35	225	20
110	110	110	110	110	110	30	220	40	20
110	30	120	120	29	123	211	225	40	30
121	30	30	30	30	28	125	125	221	123
150	30	30	30	30	111	111	111	123	123
150	150	123	123	123	150	150	151	143	123



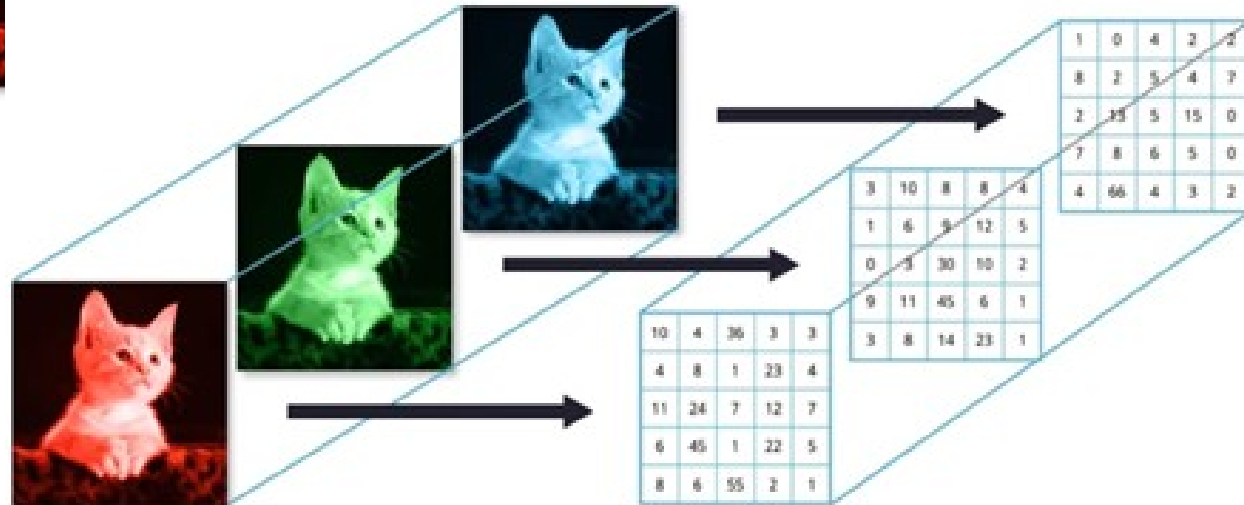
Basic Definitions

RGB Image

5 x 5 x 3

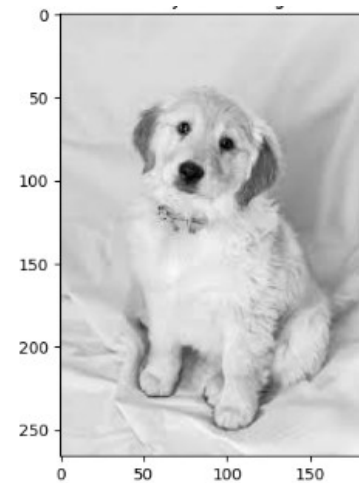
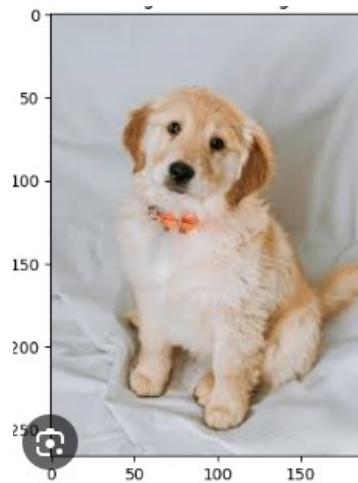


RGB Image



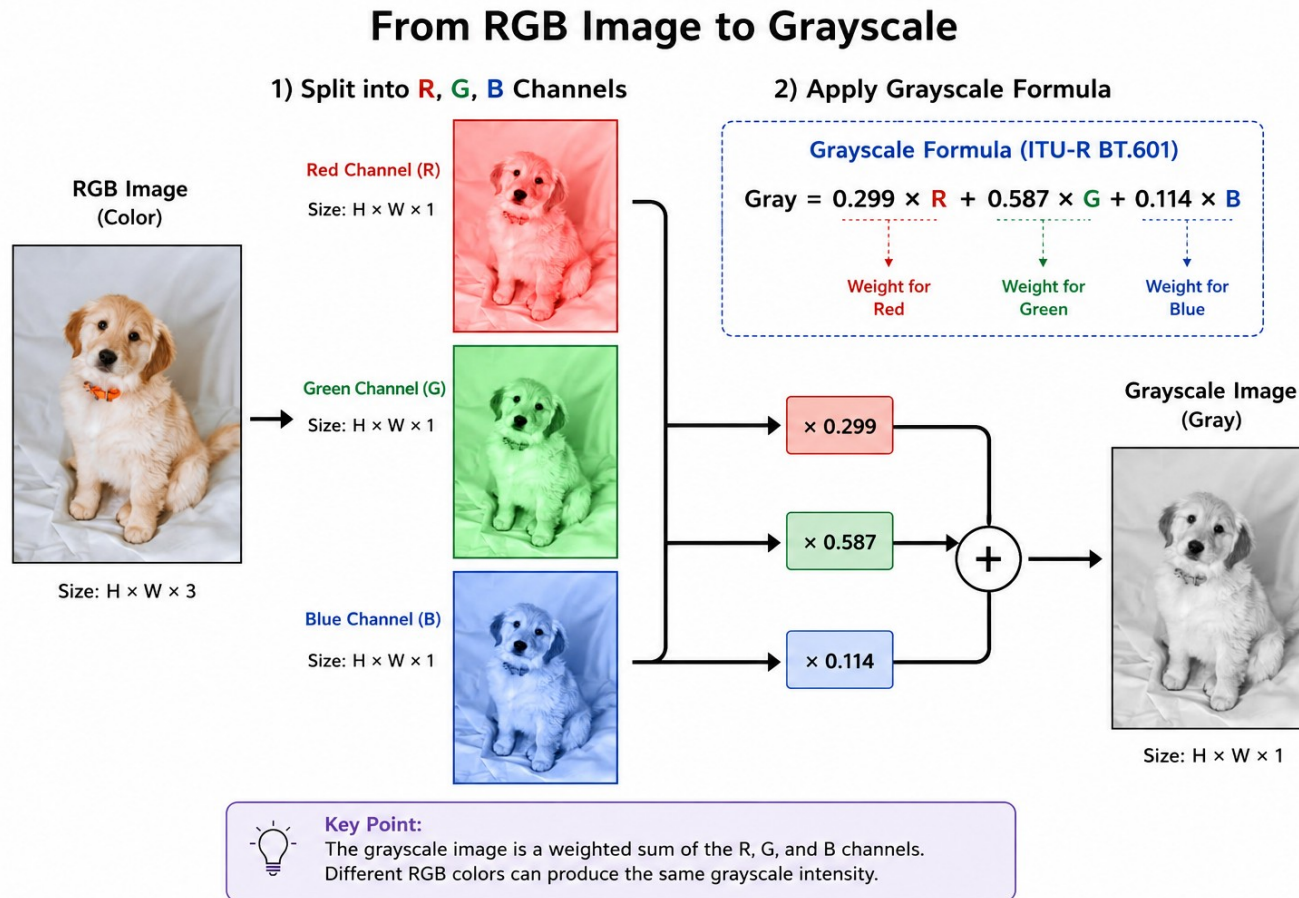
Basic Definitions

From RGB to Grayscale



Basic Definitions

From RGB to Grayscale

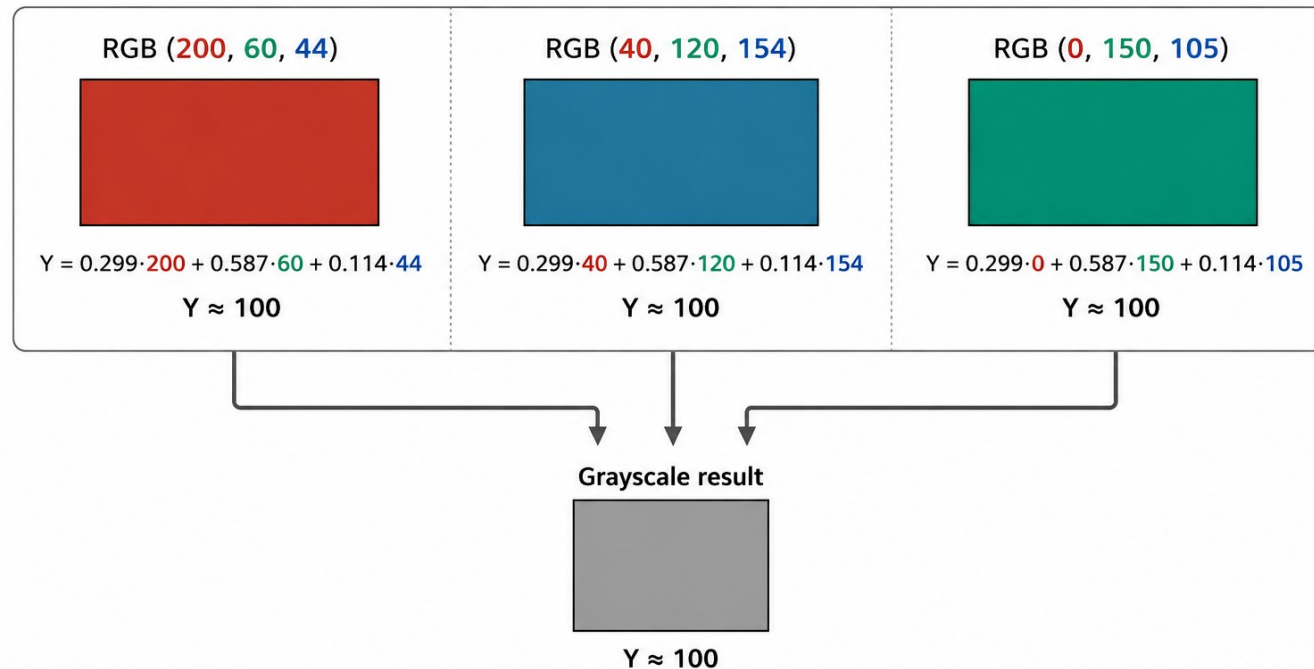


Basic Definitions

From RGB to Grayscale

Different RGB Colors → Same Grayscale

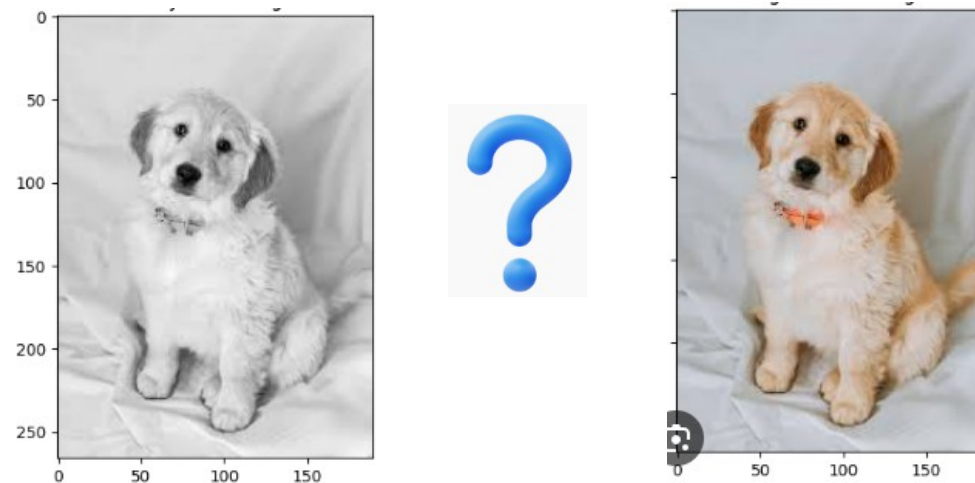
Grayscale conversion (ITU-R BT.601): $Y = 0.299R + 0.587G + 0.114B$



Multiple different RGB colors can produce the same grayscale intensity.
Therefore, the original colors cannot be uniquely recovered from a grayscale image.

Basic Definitions

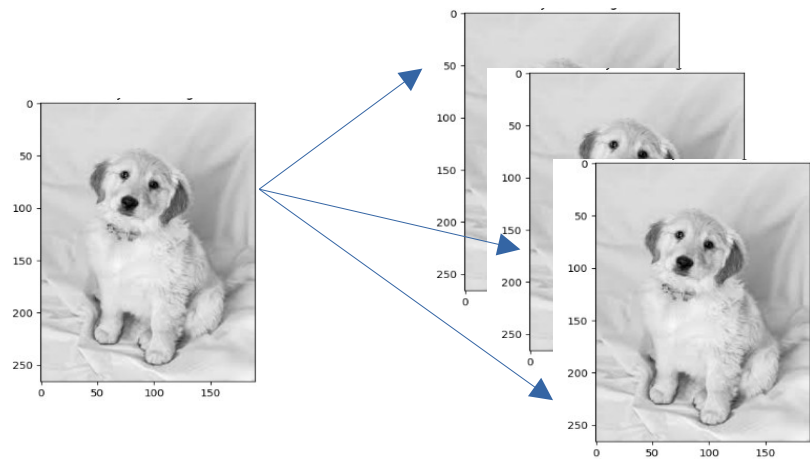
Can We Convert Grayscale to RGB?



Basic Definitions

Can We Convert Grayscale to RGB?

We can replicate the grayscale channel into the R, G, and B channels to obtain an 3-Channel image, but the original color information is irreversibly lost.

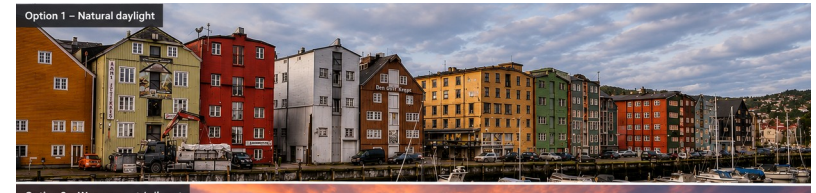
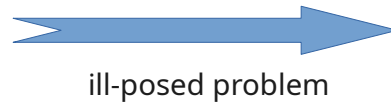


$$R = G = B$$

Basic Definitions

Recovering the original colors from a grayscale image is an ill-posed problem:

- * Multiple color images can produce the same grayscale image.
- * Image colorization algorithms estimate plausible colors rather than recovering the true ones (i.e Generative AI) .



Basic Definitions

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ill-posed problem



Basic Definitions

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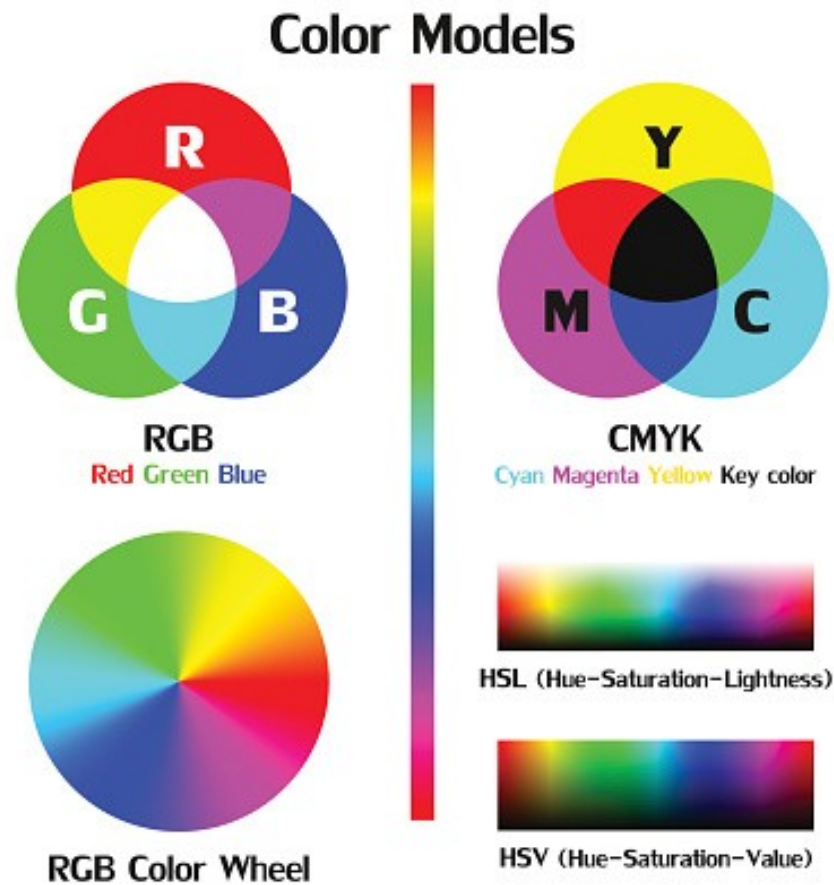


ill-posed problem



Basic Definitions

Others Color Spaces



Practice 01

Let's Code!!

<https://github.com/andrehochuli/teaching/tree/main/ComputerVision>

Lecture 01